Accuracy in Biometric Systems

FMR, FNMR, FTR & Derived Metrics

Introduction to Biometric Accuracy

- **Why Accuracy Matters:**
  - No biometric system is 100% accurate
  - Errors occur due to sensor noise, environmental factors, and biological variations
  - We need standardized metrics to measure and compare system performance
  - Different applications require different accuracy levels

Types of Errors

Type I Error: False Match

System accepts an impostor

Type II Error: False Non-Match

System rejects a genuine user

Trade-offs

Reducing one error typically increases the other

System designers must balance security vs. convenience

Threshold adjustment controls this trade-off

Basic Terminology: Match Attempts

- **Genuine Attempt**
  - Comparison between samples from the **same person**
  - **Example:** User's fingerprint compared against their own stored template
  - **Expected Result:** Should MATCH

- **Impostor Attempt**
  - Comparison between samples from **different people**
  - **Example:** Attacker's fingerprint compared against victim's template
  - **Expected Result:** Should

Match Score

Numerical value indicating similarity between two biometric samples

- **High score:** Samples are similar (likely same person)
- **Low score:** Samples are different (likely different people)
- **Threshold (τ):** Decision boundary for match/non-match

Confusion Matrix for Biometrics

- The confusion matrix shows all possible outcomes of biometric matching:

	Predicted: Match	Predicted: Non-Match
Actual: Genuine	**True Match (TM)** Correct acceptance	**False Non-Match (FNM)** Incorrect rejection
Actual: Impostor	**False Match (FM)** Incorrect acceptance	**True Non-Match (TNM)** Correct rejection

Key Insight: Green cells are correct decisions, Red/Yellow cells are errors that we want to minimize.

False Match Rate (FMR)

- Probability that the system **incorrectly accepts an impostor**

```
FMR = (Number of False Matches) /
  (Number of Impostor Attempts)
```

```
FMR = FM / (FM + TNM)
```

- **Understanding FMR**
  - **Also known as:** False Accept Rate (FAR), Type I Error
  - **Security Metric:** Lower FMR = Higher security
  - **Risk:** Unauthorized access to system/resources
  - **Critical for:** High-security applications (banking, military, border control)

- **Example:** If FMR = 0.001 (0.1%), it means 1 out of every 1000 impostor attempts will be incorrectly accepted.

False Non-Match Rate (FNMR)

- Probability that the system **incorrectly rejects a genuine user**

```
FNMR = (Number of False Non-Matches) /
       (Number of Genuine Attempts)
```

```
FNMR = FNM / (FNM + TM)
```

- **Understanding FNMR**
  - **Also known as:** False Reject Rate (FRR), Type II Error
  - **Usability Metric:** Lower FNMR = Better user experience
  - **Impact:** User frustration, denied access to legitimate users
  - **Critical for:** Convenience applications (phone unlock, attendance systems)
- **Example:** If FNMR = 0.02 (2%), it means 2 out of every 100 genuine user attempts will be incorrectly rejected.

FMR vs FNMR: The Trade-off

Threshold Setting	FMR	FNMR	Use Case
Low Threshold (Easy to match)	**High ↑**	**Low ↓**	Gym access, low-security areas
Medium Threshold (Balanced)	**Medium**	**Medium**	Office access, mobile phones
High Threshold (Hard to match)	**Low ↓**	**High ↑**	Bank vaults, military facilities

True Match Rate (TMR)

- Probability that the system **correctly accepts a genuine**

```
TMR = (Number of True Matches) /
   (Number of Genuine Attempts)
```

```
TMR = TM / (TM + FNM) = 1 - FNMR
```

- **Other Names**
  - **True Accept Rate (TAR)**
  - **Genuine Accept Rate (GAR)**
  - **Sensitivity** (in general classification)
  - **Hit Rate**
- **Important Relationship:**
  TMR + FNMR = 1
  If TMR = 0.98 (98%), then FNMR = 0.02 (2%)

True Non-Match Rate (TNMR)

- Probability that the system **correctly rejects an impostor**

```
TNMR = (Number of True Non-Matches) /
     (Number of Impostor Attempts)
```

```
TNMR = TNM / (TNM + FM) = 1 - FMR
```

- **Other Names**
  - **True Reject Rate (TRR)**
  - **Specificity** (in general classification)
- **Important Relationship:**
  TNMR + FMR = 1
  If TNMR = 0.999 (99.9%), then FMR = 0.001 (0.1%)

Summary of Basic Metrics

Metric	Formula	What it Measures	Desired Value
FMR (FAR)	FM / (FM + TNM)	Impostor acceptance rate	**Low (near 0)**
FNMR (FRR)	FNM / (FNM + TM)	Genuine rejection rate	**Low (near 0)**
TMR (TAR/GAR)	TM / (TM + FNM) = 1 - FNMR	Genuine acceptance rate	**High (near 1)**
TNMR (TRR)	TNM / (TNM + FM) = 1 - FMR	Impostor rejection rate	**High (near 1)**

Remember: You cannot minimize both FMR and FNMR simultaneously. System design involves finding the optimal balance.

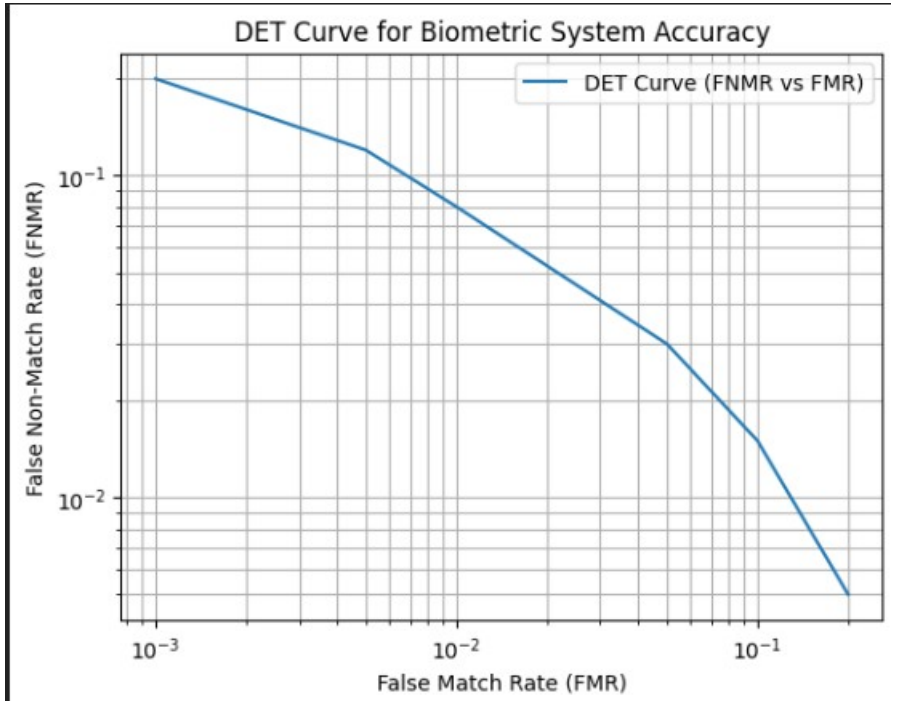
Equal Error Rate (EER)

- The point where **FMR = FNMR**

- **Why EER is Important**
  - **Single Number Summary:** Provides one metric to compare systems
  - **Industry Benchmark:** Widely used in academic papers and product specifications
  - **Lower is Better:** EER of 1% is better than EER of 5%
  - **Balance Point:** Represents balanced security and usability

Typical EER Values

Biometric Type	Typical EER	Quality
Iris Recognition	< 0.01%	**Excellent**
Fingerprint	1-2%	**Very Good**
Face Recognition	2-5%	**Good**
Voice Recognition	5-10%	**Moderate**


DET Curve for Biometric System Accuracy
DET Curve (FNMR vs FMR)
False Non-Match Rate (FNMR)
10^{-1}
10^{-2}
10^{-3}
10^{-2}
10^{-1}
False Match Rate (FMR)